Module 3: Probability — Study Notes

1. Random Phenomena

A **random phenomenon** is an experiment whose outcome **cannot be predicted with certainty** before it occurs, but in the long run the outcomes follow a predictable pattern.

Coin Toss

Possible outcomes:

```
S = {Head, Tail}
```

After many tosses:

```
P(Head) ≈ 0.5 P(Tail) ≈ 0.5
```

Rolling a Die

```
S = {1, 2, 3, 4, 5, 6}
```

```
P(each outcome) = 1/6
```

2. Sample Space

The **sample space (S)** is the set of **all possible outcomes** of a random experiment.

Example: Tossing Two Coins

```
S = {HH, HT, TH, TT}
```

Each outcome has probability:

```
P(each outcome) = 1/4
```

3. Events

An **event** is any **subset of the sample space**. If the outcome belongs to the event, the event **occurs**.

Example:

```
S = {1, 2, 3, 4, 5, 6} Event E = even numbers E = {2, 4, 6}
```

4. Special Events

1. Null Event (∅)

Event with **no outcomes**. Example: Getting number 7 on a die → impossible.

```
∅ = {}
```

2. Entire Event (S)

Event containing **all possible outcomes**. This event **always occurs**.

```
S = {all outcomes}
```

5. Set Operations on Events

Union of Events (A ∪ B)

Union means **A OR B occurs**.

```
A = even numbers = {2, 4, 6} B = numbers > 4 = {5, 6} A ∪ B = {2, 4, 5, 6}
```

Intersection of Events (A ∩ B)

Intersection means **A AND B occur together**.

```
A = {2, 4, 6} B = {4, 5, 6} A ∩ B = {4, 6}
```

Complement of an Event (A')

Complement means the **event does NOT occur**.

```
S = {1, 2, 3, 4, 5, 6} A = {2, 4, 6} A' = {1, 3, 5}
```

6. Mutually Exclusive Events

Two events are **mutually exclusive** if they **cannot occur at the same time**.

```
Condition: A ∩ B = ∅
```

Example:

```
A = even numbers = {2, 4, 6} B = odd numbers = {1, 3, 5} A ∩ B = ∅ → Mutually Exclusive
```

7. Probability of an Event

If all outcomes are equally likely:

```
P(E) = n(E) / n(S)
```

Where **n(E)** = number of favourable outcomes and **n(S)** = total number of outcomes.

Example — Rolling a Die:

```
S = {1, 2, 3, 4, 5, 6} E = {2, 4, 6} P(E) = 3 / 6 = 1/2
```

8. Basic Probability Rules

Addition Rule — Mutually Exclusive Events

```
P(A ∪ B) = P(A) + P(B)
```

Example:

```
P(A) = 0.2, P(B) = 0.3 P(A ∪ B) = 0.2 + 0.3 = 0.5
```

General Addition Rule

If events **can** occur together:

```
P(A ∪ B) = P(A) + P(B) − P(A ∩ B)
```

Example:

```
P(A) = 0.20, P(B) = 0.35, P(A ∩ B) = 0.08 P(A ∪ B) = 0.20 + 0.35 − 0.08 = 0.47
```

Complement Rule

```
P(A') = 1 − P(A)
```

Example:

```
P(pass) = 0.8 P(fail) = 1 − 0.8 = 0.2
```

9. Counting Techniques (Basic Combinatorics)

1. Addition Rule of Counting

If events are mutually exclusive:

```
n(A1 ∪ A2) = n(A1) + n(A2)
```

Example: A student can choose from 3 science clubs or 2 arts clubs → Total = 3 + 2 = 5

2. Multiplication Rule

If operations occur **in sequence**:

```
Total ways = n1 × n2
```

Example — Selecting chairperson and vice-chairperson from 10 people:

```
Step 1 (chairperson): 10 ways Step 2 (vice-chairperson): 9 ways Total = 10 × 9 = 90
```

Summary — Module 3

Topic	Concept
Random Phenomena	Experiments with uncertain outcomes
Sample Space	Set of all possible outcomes
Events	Subset of sample space
Special Events	Null event, entire event
Set Operations	Union, intersection, complement
Mutually Exclusive	Events cannot occur together
Probability Formula	P(E) = n(E) / n(S)

Probability Rules	Addition rule, complement rule
Counting Techniques	Addition rule, multiplication rule